

Transcendental values of class group L -functions

M. Ram Murty · V. Kumar Murty

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Abstract Let K be an algebraic number field and f a complex-valued function on the ideal class group of K . Then, f extends in a natural way to the set of all non-zero ideals of the ring of integers of K and we can consider the Dirichlet series $L(s, f) = \sum_{\mathfrak{a}} f(\mathfrak{a}) \mathbf{N}(\mathfrak{a})^{-s}$ which converges for $\Re(s) > 1$. After extending this function to $\Re(s) = 1$, and in the case that f does not contain the trivial character, we study the special value $L(1, f)$ when f is algebraic valued and K is an imaginary quadratic field. Applying Kronecker's limit formula and Baker's theory of linear forms in logarithms, we derive a variety of results related to the transcendence of this special value.

1 Introduction

Let K be an algebraic number field, f a $\mathbb{C}$ -valued function of the ideal class group $\mathcal{H}_K$ of K . We consider the Dirichlet series

$$L(s, f) := \sum_{\mathfrak{a}} \frac{f(\mathfrak{a})}{\mathbf{N}(\mathfrak{a})^s}, \quad (1)$$

where the summation is over all integral ideals $\mathfrak{a}$ of the ring of integers, $\mathcal{O}_K$, of K . If f is identically 1, then $L(s, f)$ is the Dedekind zeta function of K . If f is a character

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M. R. Murty (✉)

Department of Mathematics, Queen's University, Kingston, ON K7L 3N6, Canada
e-mail: murty@mast.queensu.ca

V. K. Murty

Department of Mathematics, University of Toronto, Toronto, ON M5S 2E4, Canada
e-mail: murty@math.toronto.edu

χ of the ideal class group $\mathcal{H}_K$ of K , then, $L(s, \chi)$ is a Hecke L -function. Our goal in this paper is to investigate special values of $L(s, f)$ at $s = 1$ in the case that K is an imaginary quadratic field and f is $\overline{\mathbb{Q}}$ -valued.

This study will reveal new aspects of the transcendental nature of $L(1, \chi)$ when χ is an ideal class character. In particular, we will show that the values $L(1, \chi)$ are linearly independent over $\overline{\mathbb{Q}}$ as χ ranges over non-trivial characters of the ideal class group modulo complex conjugation.

We first prove:

Theorem 1 $L(s, f)$ extends analytically for all $s \in \mathbb{C}$ except possibly at $s = 1$ where it has a simple pole with residue

$$\rho_f := \sum_{\mathfrak{a} \in \mathcal{H}_K} f(\mathfrak{a}).$$

The series (1) converges at $s = 1$ if and only if $\rho_f = 0$.

Thus, in the case that the series converges at $s = 1$, it makes sense to consider $L(1, f)$. By a deeper analysis, we will show:

Theorem 2 Let K be an imaginary quadratic field and $f : \mathcal{H}_K \rightarrow \overline{\mathbb{Q}}$ be not identically zero. Suppose that $\rho_f = 0$. Then, $L(1, f) \neq 0$ unless $f(\mathfrak{C}) + f(\mathfrak{C}^{-1}) = 0$ for every ideal class $\mathfrak{C} \in \mathcal{H}_K$. Moreover, $L(1, f)/\pi$ is a $\overline{\mathbb{Q}}$ -linear combination of logarithms of algebraic numbers. In particular, $L(1, f)/\pi$ is transcendental whenever $L(1, f) \neq 0$.

This result has several interesting corollaries.

Corollary 3 Let K be an imaginary quadratic field and χ a non-trivial character of the ideal class group of K . Then, $L(1, \chi)/\pi$ is a non-zero $\overline{\mathbb{Q}}$ -linear combination of logarithms of algebraic numbers and hence transcendental.

Since complex conjugation acts on the group of ideal class characters we see by a simple calculation that $L(1, \chi) = L(1, \overline{\chi})$ for any ideal class character χ . We denote by $\mathcal{H}_K^*$ a set of orbit representatives under this action.

Corollary 4 Let K be an imaginary quadratic field and $\mathcal{H}_K$ its ideal class group. The values $L(1, \chi)$ (as χ ranges over the non-trivial characters of $\mathcal{H}_K^*$) and π are linearly independent over $\overline{\mathbb{Q}}$.

Thus, in the special case that the ideal class group $\mathcal{H}_K$ is an elementary abelian 2-group, the corollary implies that the $L(1, \chi)$ as χ ranges over the non-trivial characters of $\mathcal{H}_K$, are linearly independent over $\overline{\mathbb{Q}}$.

In the general case, writing

$$f = \sum_{\chi \neq 1} c_\chi \chi,$$

we easily see that the condition $f(\mathfrak{C}) + f(\mathfrak{C}^{-1})$ for every class $\mathfrak{C}$ implies that $c_\chi + c_{\overline{\chi}} = 0$ for every $\chi \neq 1$. The linear independence of the $L(1, \chi)$ over $\overline{\mathbb{Q}}$ as χ ranges over the non-trivial characters of $\mathcal{H}_K^*$ is then clear from Theorem 2. To deduce linear independence along with π , we proceed as follows. Suppose that the $L(1, \chi)$ and π are linearly dependent over $\overline{\mathbb{Q}}$:

$$\sum_{\chi \in \mathcal{H}_K^*} \alpha_\chi L(1, \chi) + \beta \pi = 0,$$

with $\alpha_\chi, \beta \in \overline{\mathbb{Q}}, \beta \neq 0$. Then, setting

$$f = -\frac{1}{\beta} \sum_{\chi \in \mathcal{H}_K^*} \alpha_\chi \chi,$$

we have $L(1, f) = \pi$. By Theorem 2, we derive a contradiction.

Theorem 2 implies that at most one such $L(1, \chi)$ is algebraic:

Corollary 5 *All of the values $L(1, \chi)$ as χ ranges over the non-trivial characters of $\mathcal{H}_K^*$, with at most one exception, are transcendental.*

The elimination of this singular possibility, in other words, the proof of transcendence of $L(1, \chi)$ for all non-trivial χ seems difficult and is related to Schanuel's conjecture. Indeed, a “weaker” version of Schanuel suffices for our purposes. This is the conjecture that logarithms of algebraic numbers which are linearly independent over $\mathbb{Q}$ are algebraically independent. Assuming the “weaker” Schanuel's conjecture, one can show the transcendence of $L(1, \chi)$ for all non-trivial χ (see remarks at end as well as [8]).

When χ is a genus character, one can relate $L(1, \chi)$ to classical Dirichlet L -functions attached to quadratic characters [18]. Utilising this connection, we will prove the following.

Theorem 6 *Let K be an imaginary quadratic field with character χ_D . Then,*

$$\exp\left(\frac{L'(1, \chi_D)}{L(1, \chi_D)} - \gamma\right)$$

and π are algebraically independent. Here γ is Euler's constant.

Thus, we have from the above theorem that

$$\exp\left(\frac{L'(1, \chi_D)}{L(1, \chi_D)} - \gamma\right)$$

is transcendental. In particular,

$$\frac{L'(1, \chi_D)}{L(1, \chi_D)} \neq \gamma,$$

for any D . More generally, we have:

Corollary 7

$$\frac{L'(1, \chi_D)}{L(1, \chi_D)} - \gamma$$

is not a $\mathbb{Q}$ -linear combination of logarithms of algebraic numbers.

From the theorem, we can also deduce the following curious corollary.

Corollary 8 *If for some D , we have $L'(1, \chi_D) = 0$, then e^γ is transcendental.*

It is unlikely that such a D exists for a variety of reasons. But this seems difficult to prove. We consider a related question, namely, the vanishing of $L'(1, \chi)$ for any Dirichlet character $\chi \pmod{q}$. In this direction, we use results of Ihara et al. [10] to prove:

Theorem 9 *For q prime, the number of $\chi \neq \chi_0 \pmod{q}$ for which $L'(1, \chi) = 0$ is $O(q^\epsilon)$ for any $\epsilon > 0$.*

This last result is very analytic in flavour and it is unlikely that one can show the non-vanishing of $L'(1, \chi)$ in general using analytic methods. Theorem 6 allows us to connect this question to special values of the Γ -function via the Chowla–Selberg formula. Indeed, our proof of Theorem 6 leads to a simple proof of the Chowla–Selberg formula which we give in Sect. 7. Naturally, this leads one to enquire about the transcendence of special values of the Γ -function. Not much is known in this context. It is well-known that $\Gamma(1/2) = \sqrt{\pi}$ is transcendental by the celebrated theorem of Lindemann. The transcendence of $\Gamma(1/3)$ and $\Gamma(1/4)$ was established by Chudnovsky [3] in 1976. Recently, Grinspan [6] and Vasilev [19] independently showed that at least two of the three numbers π , $\Gamma(1/5)$, $\Gamma(2/5)$ are algebraically independent. Very likely, all of the three numbers are algebraically independent. Apart from these results, no further results are known regarding the transcendence of the Γ -function at rational, non-integral arguments. Thus, in this context, the following theorem is of interest.

Theorem 10 *Let $q > 1$ and $q|24$. Let a be coprime to q . There exists a finite set S and a collection of pair-wise non-isogenous CM elliptic curves E_j , $j \in S$ defined over $\overline{\mathbb{Q}}$ with fundamental real periods ω_j such that $\Gamma(a/q)$ lies in the field generated over $\mathbb{Q}$ by π and the ω_j . In particular, if π and the ω_j 's are algebraically independent, then $\Gamma(a/q)$ is transcendental.*

The key point here is that the non-trivial Dirichlet characters $\pmod{24}$ are all quadratic. Consequently, one can use the Chowla–Selberg formula (as we have stated it below) to express $\Gamma(a/q)$ as a product of π and periods of various non-isogenous elliptic curves.

Before moving on in our discussion, we observe an amusing corollary of the above theorem:

Corollary 11 *All of the numbers*

$$\Gamma(1/8), \Gamma(3/8), \Gamma(5/8), \Gamma(7/8)$$

are transcendental with at most one exception.

Schanuel's conjecture predicts that if $x_1, \dots, x_n$ are linearly independent over $\mathbb{Q}$, then the transcendence degree of the field

$$\mathbb{Q}(x_1, \dots, x_n, e^{x_1}, \dots, e^{x_n})$$

is at least n . The previous theorem motivates the following extension of Schanuel's conjecture. Suppose that $x_1, \dots, x_n$ are linearly independent over $\overline{\mathbb{Q}}$. Let $\wp_2, \dots, \wp_n$ be the Weierstrass $\wp$ -functions attached to non-isogenous CM elliptic curves $E_2, \dots, E_n$ defined over $\overline{\mathbb{Q}}$. If $x_2, \dots, x_n$ are not contained in the poles of the $\wp_i$, $2 \leq i \leq n$, then, the transcendence degree of the field

$$\mathbb{Q}(x_1, \dots, x_n, e^{x_1}, \wp_2(x_2), \dots, \wp_n(x_n))$$

is at least n . Thus, choosing $x_1 = \pi i$ and $x_j = \omega_j/2$ with the ω_j as in Theorem 10, the conjecture allows us to deduce that π and the ω_j 's are algebraically independent.

Our conjecture is a special case of a more general conjecture of Grothendieck (see [5]). This conjecture asserts that the transcendence degree of the field generated by the periods of an algebraic variety is equal to d where d is the dimension of the Hodge group of the variety. In our case, we consider the variety

$$X = \mathbb{P}^1 \times E_2 \times \dots \times E_n$$

where E_i are pairwise non-isogenous elliptic curves with complex multiplication. The Hodge group of $H^2(\mathbb{P}^1) \otimes \dots \otimes H^1(E_n)$ is isomorphic to

$$\mathbb{G}_m^1 \times \prod_{i=2}^n (R_{K_i/\mathbb{Q}} \mathbb{G}_m)^1,$$

where K_i is the imaginary quadratic field corresponding to E_i and the superscript denotes elements of norm 1. The dimension of this group is n .

The study of $L(s, f)$ allows us to give a simple proof of the Chowla–Selberg formula [2]. This then facilitates an application of Nesterenko's theorem to deduce the transcendence result indicated in Theorem 6.

One could discuss a more general situation where we consider functions f defined on ray class groups and similar formulas and results can be derived. However, these derivations are a bit more complicated and their treatment would alter the elegance and simplicity of this paper. Therefore, we have decided to treat the general case in a future paper.

2 Preliminaries

In the discussion below, a pivotal role will be played by the fundamental theorem of Baker concerning linear forms in logarithms. Let us note that here and later, we interpret $\log$ as the principal value of the logarithm with the argument lying in the interval $(-\pi, \pi]$. We record Baker's theorem below.

Lemma 12 *If $\alpha_1, \dots, \alpha_n \in \overline{\mathbb{Q}} \setminus \{0\}$ and $\beta_1, \dots, \beta_n \in \overline{\mathbb{Q}}$, then*

$$\beta_1 \log \alpha_1 + \dots + \beta_n \log \alpha_n$$

is either zero or transcendental. The latter case arises if $\log \alpha_1, \dots, \log \alpha_n$ are linearly independent over $\mathbb{Q}$ and $\beta_1, \dots, \beta_n$ are not all zero.

Proof This is the content of Theorems 2.1 and 2.2 of [1]. □

We will also use Kronecker's limit formula as discussed in the works of Siegel [18], Ramachandra [16] and Lang [11]. We begin by reviewing this.

Let $\Delta(z)$ be Ramanujan's cusp form:

$$\Delta(z) = q \prod_{n=1}^{\infty} (1 - q^n)^{24}, \quad q = e^{2\pi iz}.$$

Now let K be an imaginary quadratic field and let $\mathfrak{b}$ be an ideal of $\mathcal{O}_K$. If $[\beta_1, \beta_2]$ is an integral basis of $\mathfrak{b}$ with $\Im(\beta_2/\beta_1) > 0$, we define

$$g(\mathfrak{b}) = (2\pi)^{-12} (\mathbf{N}(\mathfrak{b}))^6 \Delta(\beta_2/\beta_1).$$

One can verify (as on page 109 of [16]) that $g(\mathfrak{b})$ is well-defined and does not depend on the choice of integral basis of $\mathfrak{b}$. In fact, by Lemma 2 of [16] (or page 280 of [11]), $g(\mathfrak{b})$ depends only on the ideal class $\mathfrak{b}$ belongs to in the ideal class group. Thus, if $\mathfrak{C}$ is an ideal class, we write $g(\mathfrak{C})$ for the common value $g(\mathfrak{b})$ as $\mathfrak{b}$ ranges over the elements of the class $\mathfrak{C}$.

Let d_K be the discriminant of K and w denote the number of roots of unity in $\mathcal{O}_K$. Writing

$$\zeta(s, \mathfrak{C}) = \sum_{\mathfrak{a} \in \mathfrak{C}} \frac{1}{\mathbf{N}(\mathfrak{a})^s},$$

for the ideal class zeta function, we have by Kronecker's limit formula

$$\zeta(s, \mathfrak{C}) = \frac{2\pi}{w\sqrt{|d_K|}} \left(\frac{1}{s-1} + 2\gamma - \log |d_K| - \frac{1}{6} \log |g(\mathfrak{C}^{-1})| \right) + O(s-1), \quad (2)$$

as $s \rightarrow 1^+$. (Note that there is a sign error in formula (2) on page 280 of [11].)

Proposition 13 *If $\mathfrak{C}_1$ and $\mathfrak{C}_2$ are ideal classes, then $g(\mathfrak{C}_1)/g(\mathfrak{C}_2)$ is an algebraic number lying in the Hilbert class field of K .*

Proof This follows immediately from Lemma 3 of [2] and is a classical result from the theory of complex multiplication. $\square$

Proposition 14 *For any ideal $\mathfrak{b}$, $g(\mathfrak{b})/g(\mathcal{O}_K)$ is an element lying in the Hilbert class field K_H of K . If $\mathfrak{p}$ is a prime ideal of K and $\sigma_{\mathfrak{p}}$ is the Frobenius automorphism in $\text{Gal}(K_H/K)$, then*

$$\sigma_{\mathfrak{p}}(g(\mathfrak{b})/g(\mathcal{O}_K)) = g(\mathfrak{p}^{-1}\mathfrak{b})/g(\mathfrak{p}^{-1}\mathcal{O}_K), \quad \overline{g(\mathfrak{b})/g(\mathcal{O}_K)} = g(\mathfrak{b}^{-1})/g(\mathcal{O}_K).$$

Proof The first part is the content of Theorem 1 on page 161 of [11]. The action of complex conjugation is easily deduced from the equation $\overline{j(\mathfrak{b})} = j(\overline{\mathfrak{b}})$ for the j -function. $\square$

In the later sections of the paper, we will make fundamental use of a result of Nesterenko [15].

Proposition 15 *For any imaginary quadratic field with discriminant D and character χ_D , the numbers π , $e^{\pi\sqrt{D}}$ and*

$$\prod_{a=1}^D \Gamma(a/D)^{\chi_D(a)},$$

are algebraically independent over $\overline{\mathbb{Q}}$.

Proof See Corollary 3.2, page 6 of [15]. $\square$

3 Proofs of Theorem 1 and Corollary 3

We first write

$$L(s, f) = \sum_{\mathfrak{C} \in \mathcal{H}_K} f(\mathfrak{C}) \zeta(s, \mathfrak{C}).$$

Each $\zeta(s, \mathfrak{C})$ extends to all $s \in \mathbb{C}$ with the exception of $s = 1$, where it has a simple pole with residue

$$\frac{2^{r_1} (2\pi)^{r_2} R_K}{w \sqrt{|d_K|}},$$

where r_1 is the number of real embeddings, $2r_2$ is the number of complex embeddings and R_K is the regulator of K . We conclude that $L(s, f)$ extends analytically to all $s \in \mathbb{C}$ apart from a simple pole at $s = 1$ with residue

$$\frac{2^{r_1} (2\pi)^{r_2} R_K}{w \sqrt{|d_K|}} \sum_{\mathfrak{C}} f(\mathfrak{C}).$$

Thus, $L(s, f)$ is analytic at $s = 1$ if and only if $\rho_f = 0$. To study the convergence of the Dirichlet series $L(s, f)$ at $s = 1$, we proceed as follows. The number of ideals with norm $\leq x$ and lying in a fixed class $\mathfrak{C}$ is well-known to be (see [12]),

$$\frac{2^{r_1}(2\pi)^{r_2}R_K}{w\sqrt{|d_K|}}x + O\left(x^{\frac{d}{d+1}}\right),$$

where d is the degree of K over $\mathbb{Q}$. Letting

$$S(x) = \sum_{\mathbf{N}(\mathfrak{a}) \leq x} f(\mathfrak{a}),$$

we have by the general technique of partial summation (see p. 17 of [13]) that

$$L(s, f) = s \int_1^\infty \frac{S(x)}{x^{s+1}} dx,$$

for $\Re(s) > 1$. Now,

$$S(x) = \sum_{\mathfrak{C} \in \mathcal{H}_K} \sum_{\mathfrak{a} \in \mathfrak{C}, \mathbf{N}(\mathfrak{a}) \leq x} f(\mathfrak{a}) = \sum_{\mathfrak{C} \in \mathcal{H}_K} f(\mathfrak{C}) \left(\frac{2^{r_1}(2\pi)^{r_2}R_K}{w\sqrt{|d_K|}}x + O\left(x^{\frac{d}{d+1}}\right) \right)$$

which is easily seen to be

$$\frac{2^{r_1}(2\pi)^{r_2}R_K\rho_f}{w\sqrt{|d_K|}}x + O\left(x^{\frac{d}{d+1}}\right).$$

Hence, by partial summation, it follows immediately that $L(s, f)$ converges at $s = 1$ if and only if $\rho_f = 0$. This completes the proof of Theorem 1.

To prove Corollary 3, we begin by noting that in the case K is an imaginary quadratic field, the formulas become simple and we can apply Kronecker's limit formula. In this situation, when the series converges, we have by (2),

$$\frac{L(1, f)}{\pi} = \frac{-1}{3w\sqrt{|d_K|}} \sum_{\mathfrak{C} \in \mathcal{H}_K} f(\mathfrak{C}) \log |g(\mathfrak{C}^{-1})|. \quad (3)$$

Now we invoke Proposition 13. Indeed, by this proposition, we have for the identity class $\mathfrak{C}_0$, that $g(\mathfrak{C}^{-1})/g(\mathfrak{C}_0)$ is algebraic. Thus, as $\rho_f = 0$, we have

$$\frac{L(1, f)}{\pi} = \frac{-1}{3w\sqrt{|d_K|}} \sum_{\mathfrak{C} \in \mathcal{H}_K} f(\mathfrak{C}) \log |g(\mathfrak{C}^{-1})/g(\mathfrak{C}_0)|, \quad (4)$$

for any fixed class $\mathfrak{C}_0$ of $\mathcal{H}_K$.

Specializing to the case $f = \chi$, where χ is a non-trivial character of the ideal class group $\mathcal{H}_K$, and using the celebrated theorem that $L(1, \chi) \neq 0$, we deduce Corollary 3 by virtue of Lemma 12. This completes the proofs.

4 Proofs of Theorem 2 and Corollaries 4 and 5

In view of (4), and Lemma 12 (Baker's theorem), the only part of Theorem 2 that remains to be proved is the non-vanishing of $L(1, f)$. To this end, we will require three lemmas.

Lemma 16 *Let K be an imaginary quadratic field and $f : \mathcal{H}_K \rightarrow \overline{\mathbb{Q}}$. Then, $L(1, f) = 0$ implies that $L(1, f^\sigma) = 0$ for any Galois automorphism σ of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$.*

Proof Equation (4) expresses $L(1, f)/\pi$ as a linear form of logarithms of algebraic numbers. Now choose a maximal set of $\mathbb{Q}$ linearly independent numbers from $\{\log |g(\mathfrak{C}^{-1})/g(\mathfrak{C}_0)| : \mathfrak{C} \in \mathcal{H}_K\}$. Denote this set by $\log \alpha_1, \dots, \log \alpha_t$. Thus,

$$\log |g(\mathfrak{C}^{-1})/g(\mathfrak{C}_0)| = \sum_{j=1}^t x(\mathfrak{C}, j) \log \alpha_j,$$

where the $x(\mathfrak{C}, j)$'s are rational numbers. Hence

$$L(1, f) = -\frac{\pi}{3w\sqrt{|d_K|}} \sum_{j=1}^t \sum_{\mathfrak{C} \in \mathcal{H}_K} f(\mathfrak{C}) x(\mathfrak{C}, j) \log \alpha_j.$$

If $L(1, f) = 0$, then an application of Baker's theorem (Lemma 12) gives that

$$\sum_{\mathfrak{C} \in \mathcal{H}_K} f(\mathfrak{C}) x(\mathfrak{C}, j) = 0, \quad j = 1, 2, \dots, t.$$

Since the $x(\mathfrak{C}, j)$'s are rational numbers, we deduce that for every Galois automorphism σ ,

$$\sum_{\mathfrak{C} \in \mathcal{H}_K} f^\sigma(\mathfrak{C}) x(\mathfrak{C}, j) = 0, \quad j = 1, 2, \dots, t.$$

Consequently, $L(1, f^\sigma) = 0$. □

The next lemma allows us to reduce the proof of Theorem 2 to the case when f is rational valued.

Lemma 17 *Let M be the algebraic number field generated by the values of f . Let $r = [M : \mathbb{Q}]$ and choose a basis $\beta_1, \dots, \beta_r$ of M over $\mathbb{Q}$ and write*

$$f(\mathfrak{C}) = \sum_{i=1}^r \beta_i f_i(\mathfrak{C}),$$

with $f_i(\mathfrak{C})$ rational. Then, $L(1, f) = 0$ implies $L(1, f_i) = 0$ for $i = 1, 2, \dots, r$.

Proof Let $M = M^{(1)}, \dots, M^{(r)}$ be the conjugate fields of M . The map $x \rightarrow x^{(j)}$ from M to $M^{(j)}$ extends to a Galois automorphism σ_j of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Thus,

$$f^{\sigma_j}(\mathfrak{C}) = \sum_{i=1}^r \beta_i^{(j)} f_i(\mathfrak{C}).$$

Clearly, the matrix $(\beta_i^{(j)})$ is invertible since $\beta_1, \dots, \beta_r$ is a basis, and we can express $f_i(\mathfrak{C})$ as a $\overline{\mathbb{Q}}$ -linear combination of the $f^{\sigma_j}(\mathfrak{C})$. By Lemma 16, we have that $L(1, f) = 0$ implies $L(1, f^{\sigma_j}) = 0$ for every j . Thus, $L(1, f_i) = 0$ for $1 \leq i \leq r$, as desired. $\square$

Lemma 18 *If f is rational-valued and $L(1, f) = 0$, then $f(\mathfrak{C}) + f(\mathfrak{C}^{-1}) = 0$ for every ideal class $\mathfrak{C}$.*

Proof If $L(1, f) = 0$, then

$$\sum_{\mathfrak{C} \in \mathcal{H}_K} f(\mathfrak{C}) \log |g(\mathfrak{C}^{-1})/g(\mathfrak{C}_0)| = 0.$$

Clearing denominators, we may suppose that f is integer-valued. Exponentiating the above expression gives

$$\prod_{\mathfrak{C} \in \mathcal{H}_K} \left| \frac{g(\mathfrak{C}^{-1})}{g(\mathfrak{C}_0)} \right|^{f(\mathfrak{C})} = 1.$$

To remove the absolute values, we square the expression and pair up $\mathfrak{C}$ with $\mathfrak{C}^{-1}$ and re-arrange it to deduce that

$$\prod_{\mathfrak{C}} \left[\frac{g(\mathfrak{C})}{g(\mathfrak{C}_0)} \right]^{f(\mathfrak{C}) + f(\mathfrak{C}^{-1})} = 1.$$

Each of the factors in the product is an algebraic number and applying Proposition 14, we see that

$$\prod_{\mathfrak{C}} \left[\frac{g(\mathfrak{p}^{-1}\mathfrak{C}^{-1})}{g(\mathfrak{p}^{-1}\mathfrak{C}_0)} \right]^{f(\mathfrak{C}) + f(\mathfrak{C}^{-1})} = 1,$$

for any prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$. Taking absolute values and then logarithms, we conclude that

$$\sum_{\mathfrak{C}} (f(\mathfrak{C}) + f(\mathfrak{C}^{-1})) \log |g(\mathfrak{p}^{-1}\mathfrak{C}^{-1})/g(\mathfrak{p}^{-1}\mathfrak{C}_0)| = 0,$$

for every prime ideal $\mathfrak{p}$. By the Chebotarev density theorem, the $\mathfrak{p}^{-1}$'s range over all elements of $\mathcal{H}_K$ as $\mathfrak{p}$ ranges over all prime ideals of $\mathcal{O}_K$.

We view these equations as a matrix equation

$$DF = 0$$

where F is the transpose of the row vector $(f(\mathfrak{C}^{-1}))_{\mathfrak{C} \in H_K}$ and D is the “Dedekind–Frobenius” matrix whose (i, j) th entry is given by $\log g(\mathfrak{C}_i^{-1} \mathfrak{C}_j) / g(\mathfrak{C}_i^{-1})$ with $\mathfrak{C}_i, \mathfrak{C}_j$ running over the elements of the ideal class group. The first column of D is the zero vector and we can re-write our matrix equation as

$$D_0 F_0 = 0$$

where F_0 is the transpose of the row vector $(f(\mathfrak{C}^{-1}))_{\mathfrak{C} \neq 1}$ and D_0 is the matrix obtained from D by deleting the row and column corresponding to the identity element. By the theory of the Dedekind–Frobenius determinant (see for example, p. 71 of [20]), the determinant of D_0 is

$$\prod_{\chi \neq 1} \left(\sum_{\mathfrak{a}} \chi(\mathfrak{a}) \log g(\mathfrak{a}^{-1}) \right) \neq 0,$$

since by formula (3), each factor is up to a non-zero scalar, $L(1, \chi)$, which is non-zero. Thus, $f(\mathfrak{C}) + f(\mathfrak{C}^{-1}) = 0$ for all $\mathfrak{C} \neq \mathfrak{C}_0$. Since

$$\sum_{\mathfrak{C}} f(\mathfrak{C}) + f(\mathfrak{C}^{-1}) = 0,$$

we deduce that $f(\mathfrak{C}_0) + f(\mathfrak{C}_0^{-1}) = 2f(\mathfrak{C}_0) = 0$ as well. This completes the proof. $\square$

The proof of Theorem 2 can now be given as follows. First, if f is rational-valued, we are done by the previous lemma. Lemma 17 allows us to reduce to the rational-valued case. This completes the proof.

Corollary 4 is also now immediate. Indeed, suppose that

$$\sum_{\chi \neq 1, \chi \in \mathcal{H}_K^*} c_\chi L(1, \chi) \in \overline{\mathbb{Q}}\pi,$$

for some $c_\chi \in \overline{\mathbb{Q}}$. Then, setting $f = \sum_{\chi \neq 1, \chi \in \mathcal{H}_K^*} c_\chi \chi$ we have $L(1, f)/\pi$ is algebraic. Since $\rho_f = 0$, we can apply Theorem 2 and deduce that f is identically zero. By the independence of characters, this means that each c_χ is zero.

Corollary 5 follows directly from Corollary 4 since two algebraic numbers are linearly dependent over $\overline{\mathbb{Q}}$.

5 Genus characters and $L'(1, \chi_D)$

As before, let K be an imaginary quadratic field with discriminant $D < 0$. Real-valued characters of the ideal class group of K are called genus characters. These characters can be extended to functions on the ideal classes of $\mathcal{O}_K$ in the obvious way. Such extended characters take on only the values $0, \pm 1$. By a classical theorem of Kronecker, they have a simple description. For each factorization $D = D_1 D_2$ with D_1, D_2 being fundamental discriminants, we define a character χ_{D_1, D_2} by setting it to be

$$\chi_{D_1, D_2}(\mathfrak{p}) = \begin{cases} \chi_{D_1}(\mathbf{N}(\mathfrak{p})) & \text{if } (\mathfrak{p}, D_1) = 1 \\ \chi_{D_2}(\mathbf{N}(\mathfrak{p})) & \text{if } (\mathfrak{p}, D_2) = 1. \end{cases}$$

One can show that this is well-defined and that it defines a character on the ideal class group. We refer the reader to page 60 of [18] for the background on genus characters. We have the Kronecker factorization formula:

$$L(s, \chi_{D_1, D_2}) = L(s, \chi_{D_1})L(s, \chi_{D_2}).$$

Corresponding to the factorization $D = 1 \cdot D$, we get

$$L(s, \chi_{1, D}) = \zeta(s)L(s, \chi_D).$$

The left hand side is $\zeta_K(s)$ and so we can write

$$\sum_{\mathfrak{C} \in \mathcal{H}_K} \zeta(s, \mathfrak{C}) = \zeta(s)L(s, \chi_D).$$

This identity could have been derived in other ways. Applying the Kronecker limit formula, and comparing the constant term in the Laurent expansion of both sides, we obtain as in [18]:

Proposition 19

$$\gamma L(1, \chi_D) + L'(1, \chi_D) = \frac{2\pi}{w\sqrt{|d_K|}} \sum_{\mathfrak{C} \in \mathcal{H}_K} \left(2\gamma - \log |d_K| - \frac{1}{6} \log |g(\mathfrak{C}^{-1})| \right).$$

Using Dirichlet's class number formula, we deduce:

Corollary 20

$$\frac{L'(1, \chi_D)}{L(1, \chi_D)} = \gamma - \log |d_K| - \frac{1}{6h} \sum_{\mathfrak{C} \in \mathcal{H}_K} \log |g(\mathfrak{C})|,$$

where h denotes the order of $\mathcal{H}_K$.

In particular, we deduce the following interesting result.

Theorem 21 *For any ideal class $\mathfrak{C}$,*

$$\frac{L'(1, \chi_D)}{L(1, \chi_D)} - \gamma + \frac{1}{6} \log |g(\mathfrak{C})|$$

is a $\overline{\mathbb{Q}}$ -linear combination of logarithms of algebraic numbers.

6 Proof of Theorem 6

We now analyze the asymptotic behaviour of the formula in Corollary 20 using the theory of Hurwitz zeta functions. As a result, we derive a simple proof of the Chowla–Selberg formula. Recall that the Hurwitz zeta function $\zeta(s, x)$ is defined by the series

$$\zeta(s, x) := \sum_{n=0}^{\infty} \frac{1}{(n+x)^s}.$$

This series converges for $\Re(s) > 1$ and Hurwitz showed how one can extend it to the entire complex plane apart from $s = 1$ where it has a simple pole with residue 1. Given a Dirichlet character $\chi \bmod q$, we can write

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = q^{-s} \sum_{a=1}^q \chi(a) \zeta(s, a/q).$$

Thus,

$$L'(s, \chi) = -(\log q) q^{-s} \sum_{a=1}^q \chi(a) \zeta(s, a/q) + q^{-s} \sum_{a=1}^q \chi(a) \zeta'(s, a/q).$$

Using the well-known formulas

$$\zeta(0, x) = \frac{1}{2} - x, \quad \zeta'(0, x) = \log(\Gamma(x)/2\pi),$$

where the differentiation is with respect to the s -variable, we deduce that

$$L(0, \chi) = \sum_{a=1}^q \chi(a) (1/2 - a/q),$$

and

$$L'(0, \chi) = -(\log q) L(0, \chi) + \sum_{a=1}^q \chi(a) \log \Gamma(a/q), \quad (5)$$

since $\sum_{a=1}^q \chi(a) = 0$. If χ is odd and primitive, $L(s, \chi)$ satisfies a functional equation of the form

$$A^s \Gamma((s+1)/2) L(s, \chi) = \varpi A^{1-s} \Gamma((2-s)/2) L(1-s, \bar{\chi}),$$

where ϖ (called the root number) is a complex number (see page 71 of [4]) and $A = \sqrt{q/\pi}$. We also recall that for quadratic characters χ , the root number ϖ is 1. We logarithmically differentiate this expression to obtain:

$$\log A + \frac{1}{2} \psi((s+1)/2) + \frac{L'}{L}(s, \chi) = -\log A - \frac{1}{2} \psi((2-s)/2) - \frac{L'}{L}(1-s, \bar{\chi}),$$

where $\psi(s)$ denotes the digamma function, which is the logarithmic derivative of the gamma function. Putting $s = 1$ into the formula, and using (see, for example, p. 301 of [14])

$$\psi(1) = -\gamma, \quad \psi(1/2) = -\gamma - 2 \log 2$$

we deduce

$$\frac{L'}{L}(1, \chi) = -2 \log A + \gamma + \log 2 - \frac{L'}{L}(0, \bar{\chi}). \quad (6)$$

Now we specialize our discussion to quadratic characters associated to an imaginary quadratic field K . Such a character is necessarily odd and if K has discriminant D , then this character, which we denote by χ_D is a primitive character modulo D . In this situation, we have from the functional equation

$$L(0, \chi_D) = 2h_D/w_D,$$

where h_D and w_D denote the class number and number of roots of unity of K . Thus, injecting formula (5) into equation (6), we get on exponentiating,

$$\exp\left(\frac{L'(1, \chi_D)}{L(1, \chi_D)} - \gamma\right) = (2D/A^2) \prod_{a=1}^D \Gamma(a/D)^{-\chi_D(a)w_D/2h_D}.$$

By Proposition 15, and the fact that $A = \sqrt{D/\pi}$, Theorem 6 is now immediate.

7 The Chowla–Selberg formula revisited

We can combine the calculation of the previous section with Corollary 20 to deduce the Chowla–Selberg formula:

$$\prod_{\mathfrak{C} \in \mathcal{H}_K} g(\mathfrak{C})^{1/3} = \left(\frac{1}{2\pi|d_K|}\right)^{2h_D} \prod_{a=1}^D \Gamma(a/D)^{w_D \chi_D(a)}.$$

We analyze the left hand side of this equation following [7]. Let E be an elliptic curve with complex multiplication by an order in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{-D})$. Formula (3) of [7] states that any period of E is up to an algebraic factor, given by the right hand side of the above equation. In other words,

$$f(\chi_D) := \prod_{a=1}^D \Gamma(a/D)^{\chi_D(a)}$$

is equal to a product of a power of π and a power of the period of the CM elliptic curve attached to the full ring of integers of $\mathbb{Q}(\sqrt{-D})$ (up to an algebraic factor).

More generally, we can define for any character $\chi \pmod{D}$,

$$f(\chi) = \prod_{a=1}^D \Gamma(a/D)^{\chi(a)}.$$

Let $q|D$ and χ be a real primitive character $\pmod{q}$. Let χ^* denote the character obtained by extending χ to residue classes $\pmod{D}$. Then, it is not hard to see that $f(\chi^*)$ is (up to a non-zero algebraic factor) equal to $f(\chi)$. Indeed, recall that

$$\Gamma(z)\Gamma(z+1/q)\cdots\Gamma(z+(q-1)/q) = q^{1/2-qz}(2\pi)^{(q-1)/2}\Gamma(qz).$$

Thus,

$$\begin{aligned} f(\chi^*) &= \prod_{a=1}^D \Gamma(a/D)^{\chi^*(a)} \\ &= \prod_{a=1}^q \left[\Gamma(a/D)\Gamma((a+q)/D)\cdots\Gamma((a+(D/q-1)q)/D) \right]^{\chi(a)} \\ &= \prod_{a=1}^q \left[\Gamma(a/q)(2\pi)^{(D/q-1)/2}(D/q)^{1/2-a/q} \right]^{\chi(a)} \\ &= f(\chi)(D/q)^{\sum_{a=1}^q (1/2-a/q)\chi(a)}. \end{aligned}$$

Since χ is a real character, the exponent of D/q is rational and so the second factor is algebraic and non-zero. Consequently, $f(\chi)$ and $f(\chi^*)$ are equal apart from a non-zero algebraic factor. We record this remark here since it will be used in the next section.

8 Proof of Theorem 10

We are now ready to prove Theorem 10. For each odd quadratic character χ_D , we have associated an imaginary quadratic extension k_D . Thus, $f(\chi)$ is defined for any odd

quadratic character. We can associate a CM elliptic curve E_D , with ring of endomorphisms isomorphic to the ring of integers of k_D . Let ω_D be the real period of E_D . The Chowla–Selberg formula expresses

$$\sum_{a=1}^D \chi_D(a) \log \Gamma(a/D)$$

as a $\mathbb{Q}$ -linear form in $\log \pi$, $\log \omega_D$ and the logarithm of a non-zero algebraic number. For any divisor q of 24, every non-trivial Dirichlet character mod q is quadratic. Noting that

$$\sum_{\chi \text{ even}} \chi(a) = \varphi(q)/2,$$

if $a \equiv \pm 1 \pmod{q}$ and zero otherwise, we deduce that

$$\sum_{\chi \text{ odd}} \chi(a) = \begin{cases} \varphi(q)/2 & \text{if } a \equiv 1 \pmod{q} \\ -\varphi(q)/2 & \text{if } a \equiv -1 \pmod{q} \\ 0 & \text{otherwise.} \end{cases}$$

By considering the combination

$$\prod_{\chi \text{ odd}} f(\chi)^{\chi(b)}$$

where the product is over odd characters $\pmod{q}$, we find

$$\prod_{\chi \text{ odd}} f(\chi)^{\chi(b)} = \prod_{a=1}^q \Gamma(a/q)^{\sum_{\chi \text{ odd}} \chi(ab)}.$$

Since for any divisor q of 24, $b^2 \equiv 1 \pmod{q}$ for any b coprime to q and so we have $ab \equiv 1 \pmod{q}$ implies $a \equiv b \pmod{q}$. Thus,

$$\sum_{\chi \text{ odd}} \chi(ab) = \begin{cases} \varphi(q)/2 & \text{if } a \equiv b \pmod{q} \\ -\varphi(q)/2 & \text{if } a \equiv -b \pmod{q} \\ 0 & \text{otherwise.} \end{cases}$$

and we deduce that

$$\Gamma(a/q) \Gamma(1 - a/q)^{-1}$$

is the product of an algebraic number, a power of π and a product of powers of periods of non-isogenous elliptic curves. On the other hand,

$$\Gamma(a/q)\Gamma(1-a/q)$$

is a product of π and an algebraic number. Thus, we deduce that $\Gamma(a/q)$ is (up to an algebraic factor) a product of a power of π and periods of non-isogenous elliptic curves. This completes the proof.

To prove Corollary 11, we suppose that at least two of the numbers, $\Gamma(a/8)$, $\Gamma(b/8)$ (say), among

$$\Gamma(1/8), \Gamma(3/8), \Gamma(5/8), \Gamma(7/8)$$

are algebraic. By the proof of the previous theorem, we can write each term as a product of powers of π and periods ω_1 and ω_2 of two non-isogenous CM elliptic curves. By taking appropriate powers of $\Gamma(a/8)$, $\Gamma(b/8)$, we deduce that their quotient, which is algebraic, is a product of powers of π and ω_1 . By a result of Chudnovsky [3], we know that π and ω_1 are algebraically independent. This completes the proof.

9 Proof of Theorem 9

As noted in Sect. 1, Ihara et al. proved the following theorem in [10].

Proposition 22 *Let $\Lambda_0(1) = 1$ and $\Lambda_0(n) = 0$ for $n > 1$. Define for $k \geq 1$,*

$$\Lambda_k(n) = \sum_{n_1 \cdots n_k = n} \Lambda(n_1) \cdots \Lambda(n_k),$$

where Λ denotes the von Mangoldt function. Set

$$\mu^{(a,b)} := \sum_{n=1}^{\infty} \frac{\Lambda_a(n)\Lambda_b(n)}{n^2}.$$

Then, for q prime and any $\epsilon > 0$

$$T_{a,b} := \sum_{\chi \neq \chi_0} P^{(a,b)} \left(\frac{L'}{L}(1, \chi) \right) = (-1)^{a+b} \mu^{(a,b)} \varphi(q) + O(q^\epsilon),$$

where $P^{(a,b)}(z) = z^a \bar{z}^b$.

Proof This is Theorem 5 of [10]. □

Remark It is easy to see that the series for $\mu^{(a,b)}$ converges. Indeed, $\Lambda(n) \leq \log n$ so that $\Lambda_k(n) \leq d_k(n)(\log n)^k$, where $d_k(n)$ denotes the number of factorizations of n as a product of k natural numbers. Consequently, $\Lambda_k(n) = O(n^\epsilon)$ for any $\epsilon > 0$.

We apply the previous proposition with $a = b = k$ and $a = b = 2k$. An application of the Cauchy–Schwarz inequality to the sum

$$\sum_{\chi \neq \chi_0} P^{(k,k)} \left(\frac{L'}{L}(1, \chi) \right)$$

shows that for any $k \geq 1$,

$$\#\{\chi \neq \chi_0 : L'(1, \chi) \neq 0\} \geq \frac{T_{k,k}^2}{T_{2k,2k}}.$$

Let us note that

$$T_{k,k}^2 = (\mu^{(k,k)})^2 \varphi(q)^2 + O(\varphi(q)q^\epsilon)$$

and that

$$(\mu^{(k,k)})^2 = \sum_{n_1, n_2} \frac{\Lambda_k(n_1)^2 \Lambda_k(n_2)^2}{n_1^2 n_2^2} = \sum_{n=1}^{\infty} \frac{(\Lambda_k^2 \star \Lambda_k^2)(n)}{n^2},$$

where

$$(f \star g)(n) := \sum_{d|n} f(d)g(n/d),$$

is the Dirichlet convolution. Now, if $d(n)$ denotes the number of divisors of n ,

$$\begin{aligned} \Lambda_{2k}(n)^2 &= (\Lambda_k \star \Lambda_k)^2(n) = \left(\sum_{d|n} \Lambda_k(d) \Lambda_k(n/d) \right)^2 \leq d(n) \sum_{d|n} \Lambda_k^2(d) \Lambda_k^2(n/d) \\ &= d(n) (\Lambda_k^2 \star \Lambda_k^2)(n), \end{aligned}$$

by an application of the Cauchy–Schwarz inequality. As $d(n) = O(n^\epsilon)$ for any $\epsilon > 0$, we obtain

$$\mu^{(2k,2k)} = \sum_{n=1}^{\infty} \frac{\Lambda_{2k}^2(n)}{n^2} \leq \sum_{n=1}^{\infty} \frac{(\Lambda_k^2 \star \Lambda_k^2)(n)}{n^{2-\epsilon}}.$$

Putting

$$G_k(s) = \sum_{n=1}^{\infty} \frac{(\Lambda_k^2 \star \Lambda_k^2)(n)}{n^s}$$

we conclude

$$\frac{T_{k,k}^2}{T_{2k,2k}} \geq \frac{G_k(2)\varphi(q)^2 + O(\varphi(q)q^{\epsilon_2})}{G_k(2 - \epsilon_1)\varphi(q) + O(q^{\epsilon_2})}$$

for any $\epsilon_1, \epsilon_2 > 0$. Choosing $k = 2$ and noting that

$$G_2(2 - \epsilon_1) = G_2(2) + O(\epsilon_1),$$

we conclude that

$$\frac{T_{k,k}^2}{T_{2k,2k}} \geq \varphi(q) + O(q^\epsilon).$$

The result immediately follows from choosing $\epsilon_1 = 1/q$.

10 Concluding remarks

It is clear from the preceding discussion that the non-vanishing of certain Dirichlet series is connected with linear independence of special values of L -series. Such a theme was explored in a classical context in [9]. What is interesting in this paper is the role played by $L'(1, \chi)$ with χ a Dirichlet character.

The question of non-vanishing of $L'(1, \chi)$ arises in other contexts like the following. Let K be an algebraic number field and $\zeta_K(s)$ its Dedekind zeta function. It is well-known that $\zeta_K(s)$ has a simple pole at $s = 1$ with residue λ_K . Here,

$$\lambda_K = \frac{2^{r_1} (2\pi)^{r_2} h_K R_K}{w \sqrt{|d_K|}},$$

where r_1 is the number of real embeddings of K and $2r_2$ is the number of non-real embeddings of K , h_K , R_K , w and d_K are the class number, regulator, the number of units of finite order and discriminant, respectively, of K . Let us set

$$g_K(s) = \zeta_K(s) - \lambda_K \zeta(s).$$

Then, $g_K(s)$ is regular at $s = 1$. In [17], Scourfield asked if for any field $K \neq \mathbb{Q}$ we have $g_K(1) = 0$. This question is really about non-vanishing of linear combinations of derivatives of L -functions.

To see this, we write

$$\zeta_K(s) = \zeta(s) F_K(s),$$

where $F_K(s)$ is a product of certain Artin L -series. Using Brauer's induction theorem and the non-vanishing of Hecke L -series at $s = 1$, it is easily seen that $F_K(s)$ is regular

at $s = 1$. Consequently, $F_K(1) = \lambda_K$ and since

$$\zeta_K(s) - \lambda_K \zeta(s) = \zeta(s)(F_K(s) - \lambda_K),$$

we see that $g_K(1) = F'_K(1)$. If $\widehat{K}$ denotes the normal closure of K over $\mathbb{Q}$, and $G = \text{Gal}(\widehat{K}/\mathbb{Q})$, one can express $F_K(s)$ as a product of Artin L -series attached to irreducible characters of G . Indeed, if $H = \text{Gal}(\widehat{K}/K)$, $\zeta_K(s)$ is the Artin L -series attached to the character $\text{Ind}_H^G 1$. If χ is an irreducible character of G , we have by Frobenius reciprocity,

$$c_\chi := (\text{Ind}_H^G 1, \chi) = (1, \chi|_H)$$

which is the multiplicity of the trivial character in χ restricted to H . Thus, c_χ is a non-negative integer and we have

$$F_K(s) = \prod_{\chi \neq 1} L(s, \chi)^{c_\chi},$$

where the product is over the non-trivial irreducible characters of G . Hence,

$$\frac{F'_K(1)}{F_K(1)} = \sum_{\chi \neq 1} c_\chi \frac{L'}{L}(1, \chi).$$

In the special case $K/\mathbb{Q}$ is Galois, $c_\chi = \chi(1)$. Thus, in the Galois case, the question of non-vanishing of $g_K(1)$ is equivalent to the non-vanishing of

$$\sum_{\chi \neq 1} \chi(1) \frac{L'}{L}(1, \chi).$$

If $K = \mathbb{Q}(\zeta_q)$ is the q th cyclotomic field, with ζ_q being a primitive q th root of unity, then Ihara et al. [10] have investigated the asymptotic behaviour of this sum. They proved that

$$\lim_{q \rightarrow \infty, q \text{ prime}} \frac{1}{\phi(q)} \sum_{\chi \neq \chi_1} \frac{L'}{L}(1, \chi) = 0.$$

So the question of non-vanishing of $g_K(1)$ is a bit delicate and cannot be deduced from this limit theorem.

The non-vanishing of $L'(1, \chi)$ seems to be intimately linked with arithmetic questions. For example, if $K/\mathbb{Q}$ is quadratic, then $F_K(s) = L(s, \chi_D)$ where χ_D is the quadratic character attached to K . In this case, Scourfield's question reduces to the question of whether $L'(1, \chi_D) = 0$ for any such χ_D . It is unlikely that such a χ_D exists.

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